

Sanket Vinod Wardhekar

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EDUCATION

Stony Brook University

PhD, Mechanical Engineering, specialization in Solid Mechanics and Finite Element Analysis

GPA: 4.0/4.0

Aug. 2021 - May 2026

Stony Brook, NY

College of Engineering Pune

Bachelor of Engineering, Mechanical Engineering, minor in Computer Engineering

GPA: 9.43/10

Jul. 2018 - Jul. 2021

Pune, India

SKILLS

Technical Expertise: Finite Element Analysis, Constitutive Modeling, Fracture Mechanics, Impact Mechanics, Sports Biomechanics, Material Testing & Characterization, Composite Material Fabrication, Product Design, Geometric Dimensioning & Tolerancing (GD&T), 3D Solid Modeling, Data Analysis & Visualization.

Programming: Python, Fortran, MATLAB, C++, SQL, LaTeX, Microsoft Office.

Software: ABAQUS, LS-DYNA, ANSYS, HyperView, HpyerMesh, Optistruct, Creo Parametric, SolidWorks, CATIA, AutoCAD, Autodesk Fusion 360, LabVIEW.

Instruments/Tools: Instron Testing Machine, Universal Testing Machine, Scanning Electron Microscope (SEM), Bench Drill Press, Vacuum Chamber, FTIR, DMA.

WORK EXPERIENCE

Finite Element Analysis Intern PING

May 2025 – Aug. 2025
Phoenix, AZ

- Developed an advanced LS-DYNA material model capturing the anisotropic (directional) behavior of titanium alloy used in golf drivers. The model improved the durability prediction with an accuracy of ~96%.
- Lead and executed a complete dynamic impact FEA model for a new, pre-launch product using Creo/HyperMesh/LS-DYNA framework guiding the design team for an optimized design that outperformed the baseline concept.

Materials and Mechanics Graduate Research Assistant Stony Brook University

Aug. 2021 – Present
Stony Brook, NY

- Developed a constitutive model for polymers in Abaqus outperforming traditional models like Drucker-Prager by 70% better accuracy in predicting material behavior.
- Led experimental characterization, fabrication of epoxy samples and conducting ASTM-standard test, along with novel fracture mechanics experiments under complex loading conditions for model validation.

Machine Learning and Optimization Intern Tata Research Development and Design Centre (TRDDC)

Aug. 2020 – Jun. 2021
Pune, India

- Developed an optimization framework in Dakota (Sandia National Lab), integrating surrogate model, FEA, genetic optimization, and uncertainty quantification to optimize the airplane wing box weight by ~20%.
- Post processing of the framework included model validation with test data, input parameters heat mapping, and computational time assessment by using Python (scikit-learn, NumPy, and Matplotlib).

Product Strategy Intern Bajaj Auto Ltd.

Jun. 2020 – Aug. 2020
Pune, India

- Conducted a competitive benchmarking study across six major two-wheeler brands to identify gaps in after sales service. Leveraged customer insights and data analysis to evaluate service operations, spare distribution, and digital reach.
- Recommended integrating test ride booking, EMI calculators, and online service scheduling, leading to an estimated 15% boost in customer engagement and faster service response time.

RESEARCH PROJECTS

Gap Test and Size Effect Analysis of Lab-Cured Epoxy Resin

Tools: Vacuum Chamber, Instron Testing Machine, Bench Drill, CNC Lathe Machine

2024 - Present

- Preparing lab-cured epoxy resin samples following ASTM standards for uniaxial tension, compression, and size effect tests.
- Characterizing material properties of lab-cured epoxy resin on Instron testing machine and performing size effect analysis on geometrically scaled samples to identify epoxy as quasibrittle material.
- Designing and performing a novel gap test on polymers to investigate influence of crack parallel stress on its fracture properties.

Microplane Constitutive Model for Polymers

Tools: Abaqus, VUMAT, MATLAB, Python, Fortran, LaTeX

2021 - 2024

- Developed a multi-scale material model using VUMAT to predict tensile and compressive damage in polymers, incorporating key damage mechanisms, with the overall goal to implementing it in composite materials.
- Implemented in ABAQUS Explicit, capturing complex polymer behaviors like tension-compression asymmetry, pressure sensitivity, barreling, and shear band localization, with validation from uniaxial and triaxial tests.

Determination of Mohr-Coulomb Model Parameters using Regression-Based Architecture

Tools: Sampling Methods, Multivariate Regression, Scikit, Python, Abaqus

2023 - 2024

- Developed a multivariate regression model to automate the calibration process of Mohr-Coulomb material model.
- Compared Random and Latin Hypercube sampling methods to ensure comprehensive coverage of model parameters.

Cohesive Zone Model for Robotic Cutting Simulation

Tools: Robotic Hand, Abaqus, Python

2021 – 2022

- Implemented a cohesive zone model to simulate the material separation process during robotic cutting operation.
- Performed implicit FEA to predict cutting force with ~82% accuracy and material behavior for the input load required by the robotic cutting hand.

PATENTS AND PUBLICATIONS [\(All papers on google scholar\)](#)

- [1] Wardhekar, S., Kirane, K. “Size Effect Characterization of Epoxy Resin and Microplane Model Validation” Manuscript submitted for publication in Elsevier *Journal Extreme Mechanics Letter*.
- [2] Wardhekar, S., and Kirane, K. (January 24, 2025). “Microplane Constitutive Model for Tension-Compression Asymmetry and Pressure-Sensitive Damage in Polymers.” ASME. *J. Appl. Mech.* March 2025; 92(3): 031002. <https://doi.org/10.1115/1.4067565>
- [3] Wardhekar, Sanket Vinod, Smith, Garret, Kirane, Kedar. (2023). Adaptation of the Microplane Constitutive Model for Brittle-Plastic Glassy Polymers. <https://doi.org/10.12783/asc38/36598>
- [4] Sanket Vinod Wardhekar, Vispi Neville Karkaria, Hanmant Shinde, et al. (2019). Modified Hot Water Supply System for Water and Energy Savings. Patent Office, Patent No. 201921021898. Registration Date: 3rd June 2021.

LEADERSHIP ROLES AND HONORS

- Resident Assistant in Chapin Apartments, Stony Brook University (May 2024 – Present)
- President, Graduate Badminton Union, Stony Brook University (May 2024 – Present)
- J N Tata Endowment Scholar (Jul. 2021 – Present)
- National Science Foundation NRT Travel and Professional Development Award (Oct. 2023 & Nov. 2024)
- 5th Rank holder in Mechanical Engineering Department, College of Engineering Pune (Aug. 2021)